

# Nộp bài tập homework ss06

|     |                 |
|-----|-----------------|
| Tên | Hoàng Duy Hương |
| Lớp | PD16            |

## Homework 1:

- Open timing report and highlight 3 logic cell have high fanout (>32) in design
  - Dùng lệnh **restoreDesign** để mở lại design:

```
innovus 15> restoreDesign SAVED/02_place_opt.invs.dat/ croc_chip
```

- Report timing của pathgroup reg2reg

```
innovus 15> report_timing -path_group reg2reg
```

- Kết quả

```
Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_67__reg/CLK
Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_67__reg/D (*) checked with leading edge of 'clk_sys'
Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_20__reg/Q (*) triggered by leading edge of 'clk_sys'
Path Groups: {reg2reg}
Analysis View: Func_View_wc
Other End Arrival Time 0.000
- Setup 0.213
+ Phase Shift 10.000
- Uncertainty 0.100
= Required Time 9.687
- Arrival Time 57.971
= Slack Time -48.284
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
```

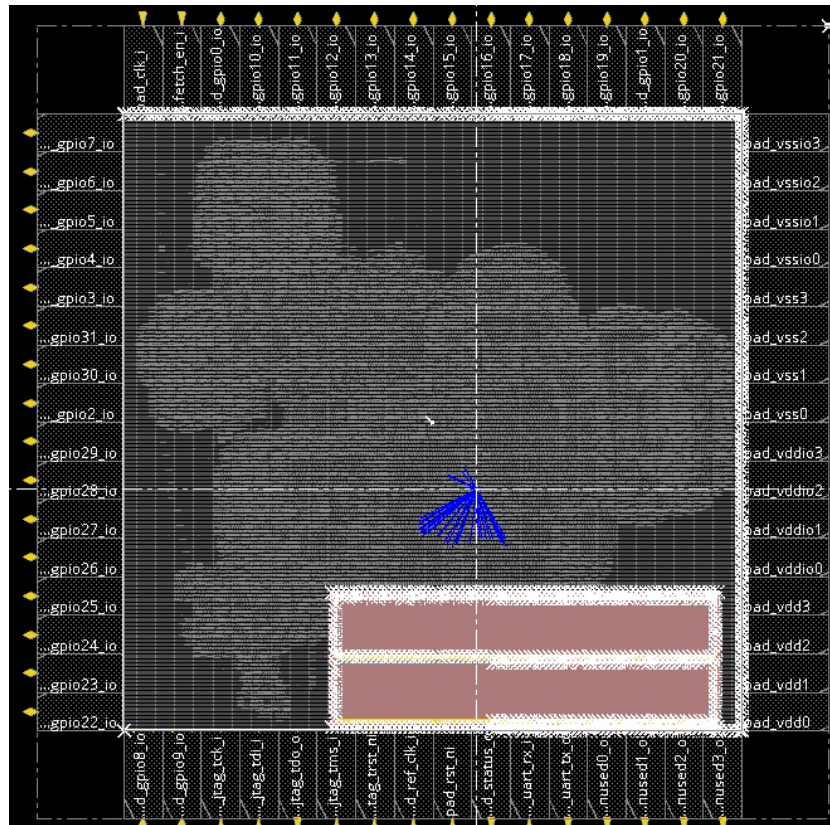
| Pin                                                                              | Cell           | Fanout | Load  | Slew   | Delay  | Arrival Time | Pin Location      |
|----------------------------------------------------------------------------------|----------------|--------|-------|--------|--------|--------------|-------------------|
| i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_20__reg/CLK | -              | 5001   | 0.000 | 0.200  | -      | 0.000        | (1164.00, 707.28) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_20__reg/Q   | sg13g2_dfrbp_1 | 434    | 2.980 | 18.635 | 11.571 | 11.571       | (1173.60, 706.44) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/_5005_Y                     | sg13g2_nor2b_2 | 28     | 0.167 | 4.865  | 7.837  | 19.408       | (1204.32, 878.64) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/_5343_Y                     | sg13g2_a22o1_1 | 1      | 0.005 | 1.730  | 0.871  | 20.878       | (1238.40, 786.24) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/_5347_Y                     | sg13g2_o21a1_1 | 1      | 0.005 | 0.285  | 0.323  | 20.402       | (1240.32, 789.36) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/_5348_Y                     | sg13g2_nand2_1 | 1      | 0.004 | 0.197  | 0.141  | 20.543       | (1244.16, 772.38) |
| i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/_5361_Y                     | sg13g2_nand3_1 | 3      | 0.063 | 0.848  | 0.746  | 21.289       | (1244.16, 768.60) |
| i_croc_soc/i_croc/i_core_wrap/_5692_Y                                            | sg13g2_o21a1_1 | 1      | 0.004 | 0.242  | 0.338  | 21.627       | (1138.56, 757.68) |
| i_croc_soc/i_croc/i_core_wrap/_5693_Y                                            | sg13g2_a21o1_1 | 12     | 0.090 | 0.866  | 0.500  | 22.127       | (1137.60, 753.48) |
| i_croc_soc/i_croc/i_core_wrap/_5698_Y                                            | sg13g2_xnor2_1 | 6      | 0.033 | 0.542  | 0.794  | 22.631       | (1023.84, 838.74) |
| i_croc_soc/i_croc/i_core_wrap/_5700_Y                                            | sg13g2_nor4_1  | 2      | 0.008 | 0.313  | 0.221  | 23.053       | (1021.92, 819.84) |
| i_croc_soc/i_croc/i_core_wrap/_5701_Y                                            | sg13g2_a21o1_1 | 2      | 0.008 | 0.511  | 0.238  | 23.290       | (1027.68, 821.62) |
| i_croc_soc/i_croc/i_core_wrap/_5722_Y                                            | sg13g2_nor2_1  | 1      | 0.004 | 0.120  | 0.162  | 23.452       | (1024.80, 819.84) |
| i_croc_soc/i_croc/i_core_wrap/_5762_Y                                            | sg13g2_a22o1_1 | 1      | 0.043 | 0.989  | 0.787  | 24.239       | (1023.36, 812.28) |
| i_croc_soc/i_croc/i_core_wrap/_6060_X                                            | sg13g2_o21o_1  | 5      | 0.023 | 0.168  | 0.599  | 24.747       | (954.24, 798.54)  |
| i_croc_soc/i_croc/i_core_wrap/_6092_X                                            | sg13g2_a21o_1  | 4      | 0.020 | 0.160  | 0.297  | 25.044       | (948.96, 851.38)  |
| i_croc_soc/i_croc/i_core_wrap/_6370_Y                                            | sg13g2_nand4_1 | 2      | 0.000 | 0.340  | 0.287  | 25.331       | (947.52, 829.60)  |
| i_croc_soc/i_croc/i_core_wrap/_6383_X                                            | sg13g2_nand3_1 | 2      | 0.009 | 0.076  | 0.268  | 25.698       | (938.40, 832.86)  |
| i_croc_soc/i_croc/i_core_wrap/_6385_X                                            | sg13g2_or2_1   | 5      | 0.001 | 0.467  | 0.515  | 26.113       | (927.36, 839.28)  |
| i_croc_soc/i_croc/_3894/Y                                                        | sg13g2_nor4_1  | 2      | 0.010 | 0.407  | 0.431  | 26.544       | (879.84, 782.04)  |
| i_croc_soc/i_croc/_3898/Y                                                        | sg13g2_nand3_1 | 3      | 0.013 | 0.266  | 0.357  | 26.901       | (880.32, 783.55)  |

- Highlight 3 logic cell có fanout lớn (> 32)

|                           |                 |    |       |       |       |        |                  |
|---------------------------|-----------------|----|-------|-------|-------|--------|------------------|
| i_croc_soc/i_croc/_3843_Y | sg13g2_nor3_1   | 2  | 0.008 | 0.231 | 0.245 | 26.034 | (911.92, 809.64) |
| i_croc_soc/i_croc/_3844_Y | sg13g2_o21a1_1  | 38 | 0.194 | 1.855 | 1.496 | 29.580 | (911.04, 682.08) |
| i_croc_soc/i_croc/_3845_X | sg13g2_and4_1   | 4  | 0.010 | 0.148 | 0.651 | 30.231 | (852.00, 706.44) |
| i_croc_soc/i_croc/_4043_Y | sg13g2_o21a1_1  | 45 | 0.188 | 2.753 | 2.076 | 32.888 | (846.72, 729.12) |
| i_croc_soc/i_croc/_4046_Y | sg13g2_nor2_1   | 1  | 0.004 | 0.415 | 0.537 | 32.844 | (844.32, 756.00) |
| i_croc_soc/i_croc/_4056_Y | sg13g2_o21a1_1  | 2  | 0.013 | 0.466 | 0.261 | 33.186 | (838.56, 757.68) |
| i_croc_soc/i_croc/_4088_Y | sg13g2_nand2b_1 | 1  | 0.004 | 0.346 | 0.241 | 33.347 | (789.12, 761.46) |
| i_croc_soc/i_croc/_4081_Y | sg13g2_o21a1_1  | 6  | 0.022 | 0.685 | 0.362 | 33.709 | (783.84, 759.36) |
| i_croc_soc/i_croc/_4184_Y | sg13g2_nor2_1   | 1  | 0.006 | 0.440 | 0.258 | 33.987 | (782.40, 757.68) |
| i_croc_soc/i_croc/_4185_Y | sg13g2_a22o1_1  | 1  | 0.004 | 0.555 | 0.193 | 34.160 | (779.52, 753.48) |
| i_croc_soc/i_croc/_4198_Y | sg13g2_o21a1_1  | 2  | 0.008 | 0.326 | 0.349 | 34.510 | (773.76, 751.80) |
| i_croc_soc/i_croc/_4228_X | sg13g2_and2_1   | 3  | 0.015 | 0.119 | 0.285 | 34.794 | (758.88, 751.80) |
| i_croc_soc/i_croc/_4222_Y | sg13g2_o21o1_2  | 6  | 0.019 | 0.142 | 0.178 | 34.969 | (759.84, 757.26) |
| i_croc_soc/i_croc/_4976_Y | sg13g2_nand2b_1 | 75 | 0.342 | 3.171 | 2.357 | 37.327 | (753.12, 761.46) |
| i_croc_soc/i_croc/_4998_Y | sg13g2_nor3_1   | 4  | 0.014 | 0.846 | 0.934 | 38.201 | (718.56, 789.60) |

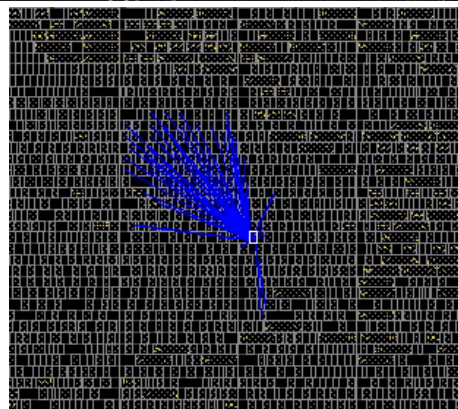
- Cell thứ nhất: **i\_croc\_soc/i\_croc/\_3844** (38 fanout)

- Cmd: selectInst i\_croc\_soc/i\_croc/\_3844



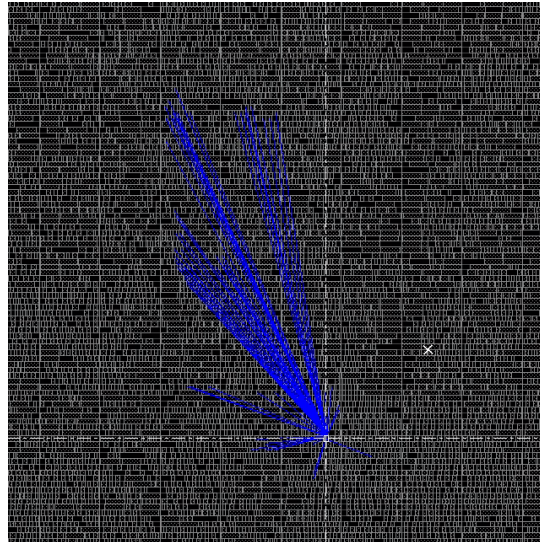
- Cell thứ 2: i\_croc\_soc/i\_croc/\_4043 (45 fanout)

|                            |               |    |       |       |       |        |                  |
|----------------------------|---------------|----|-------|-------|-------|--------|------------------|
| i_croc_soc/i_croc/_3843 JV | sg13g2_nor3_1 | 2  | 0.008 | 0.113 | 0.245 | 29.864 | (913.92, 889.64) |
| i_croc_soc/i_croc/_3844 JV | sg13g2_nor1_1 | 38 | 0.104 | 1.855 | 1.496 | 29.580 | (913.04, 652.08) |
| i_croc_soc/i_croc/_3845 JV | sg13g2_nor4_1 | 4  | 0.018 | 0.148 | 0.651 | 30.231 | (852.80, 766.44) |
| i_croc_soc/i_croc/_4043 JV | sg13g2_nor1_1 | 45 | 0.189 | 2.753 | 2.875 | 32.488 | (746.32, 720.32) |
| i_croc_soc/i_croc/_4049 JV | sg13g2_nor3_1 | 1  | 0.008 | 0.415 | 0.537 | 32.844 | (844.32, 750.60) |
| i_croc_soc/i_croc/_4050 JV | sg13g2_nor1_1 | 2  | 0.013 | 0.460 | 0.261 | 33.109 | (838.56, 757.68) |
| i_croc_soc/i_croc/_4080 JV | sg13g2_nor2_1 | 1  | 0.008 | 0.346 | 0.241 | 33.247 | (789.32, 751.46) |
| i_croc_soc/i_croc/_4081 JV | sg13g2_nor1_1 | 6  | 0.022 | 0.685 | 0.362 | 33.799 | (783.84, 759.35) |
| i_croc_soc/i_croc/_4184 JV | sg13g2_nor2_1 | 1  | 0.006 | 0.440 | 0.258 | 33.957 | (782.40, 757.68) |
| i_croc_soc/i_croc/_4185 JV | sg13g2_nor1_1 | 1  | 0.004 | 0.556 | 0.193 | 34.180 | (779.52, 765.46) |
| i_croc_soc/i_croc/_4186 JV | sg13g2_nor1_1 | 2  | 0.008 | 0.358 | 0.349 | 34.519 | (773.28, 751.46) |
| i_croc_soc/i_croc/_4220 JV | sg13g2_nor2_1 | 3  | 0.019 | 0.119 | 0.265 | 34.794 | (758.88, 751.60) |
| i_croc_soc/i_croc/_4222 JV | sg13g2_nor1_2 | 6  | 0.019 | 0.142 | 0.175 | 34.899 | (759.84, 757.36) |
| i_croc_soc/i_croc/_4216 JV | sg13g2_nor2_1 | 16 | 0.345 | 0.113 | 2.333 | 37.871 | (750.32, 761.46) |



- Cell thứ 3: **i\_croc\_soc/i\_croc/\_4976** (75 fanout)

|                            |                |    |       |       |       |        |                  |
|----------------------------|----------------|----|-------|-------|-------|--------|------------------|
| i_croc_soc/i_croc/_4185 JV | gl3g2_a22o1_1  | 1  | 0.004 | 0.555 | 0.193 | 84.160 | (779.52, 753.48) |
| i_croc_soc/i_croc/_4198 JV | gl3g2_a21a1_1  | 2  | 0.008 | 0.326 | 0.149 | 34.510 | (773.76, 751.89) |
| i_croc_soc/i_croc/_4220 JV | gl1g2_ae02_1   | 3  | 0.015 | 0.119 | 0.259 | 34.794 | (758.80, 751.60) |
| i_croc_soc/i_croc/_4222 JV | gl1g2_a21o1_2  | 6  | 0.019 | 0.142 | 0.175 | 34.969 | (759.84, 757.26) |
| i_croc_soc/i_croc/_4976 JV | gl1g2_aandib_1 | 75 | 0.342 | 3.171 | 2.357 | 37.327 | (753.12, 761.46) |
| i_croc_soc/i_croc/_4998 JV | gl1g2_mor1_1   | 4  | 0.014 | 0.046 | 0.324 | 36.261 | (716.56, 789.69) |
| i_croc_soc/i_croc/_4998 JV | gl1g2_mor1_1   | 2  | 0.008 | 0.235 | 0.301 | 38.662 | (713.76, 787.68) |



- Check design utilization by CheckPlace command

```

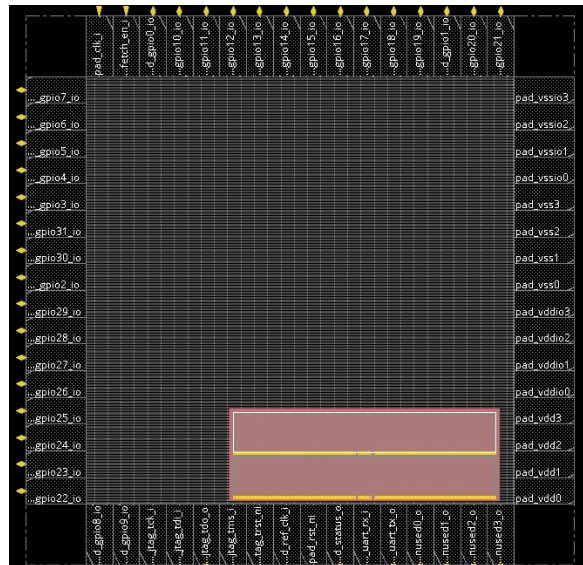
innovus 25> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=1301.5M, init mem=1337.1M)
Pre-route DRC Violation:      228
*info: Placed = 40240          (Fixed = 10253)
*info: Unplaced = 0
Placement Density:38.98%(542379/1391267)
Placement Density (including fixed std cells):39.91%(563774/1412663)
Finished checkPlace (total: cpu=0:00:01.5, real=0:00:01.0; vio checks: cpu=0:00:00.9, real=0:00:01.0; mem=1337.1M)
1
innovus 26>

```

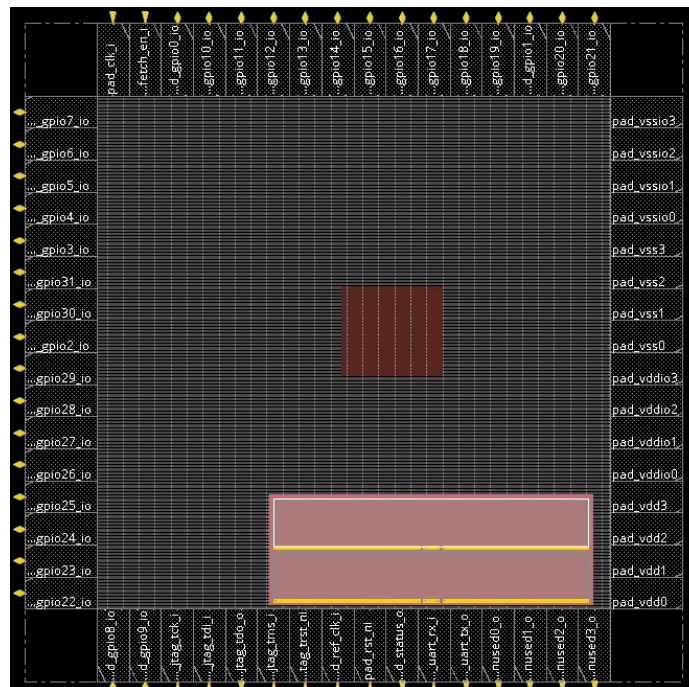
- Utilization của design hiện tại là 38.98%
- Utilization của design nếu tính luôn các fixed standard cell là **39.91%**
- Khi chọn để báo cáo thì chọn cái thứ 2 design bao gồm các std cell đã có fixed!

## Homework 2:

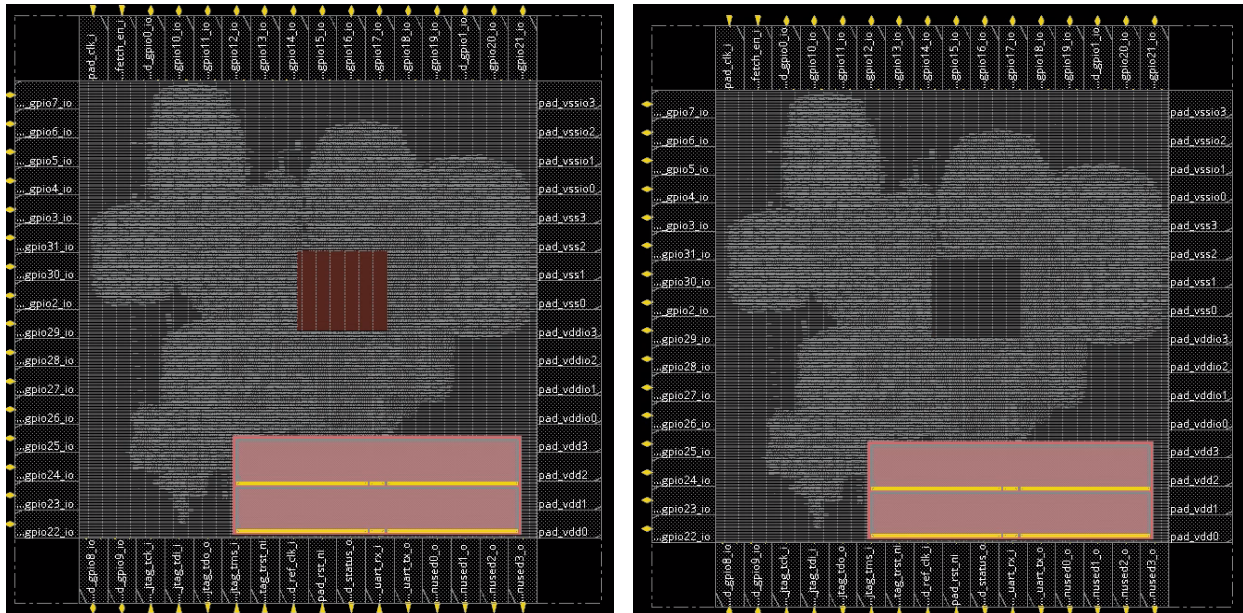
- Source script : `source ../pr/script/00_init_design.tcl`



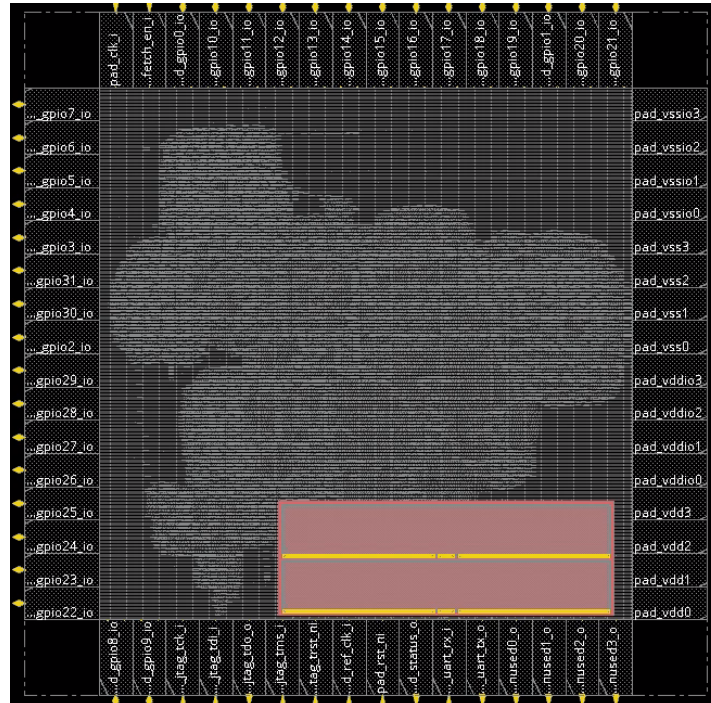
- Apply add hard blockage and run place design follow below command
  - `createPlaceBlockage -type hard -box {791.04 762.3 1039.2 985.32}`



## ○ Place\_design



- Các cell sẽ đặt quanh hard blockage, và không được phép đặt vào hard blockage
- Design khi không sử dụng hard blockage



- Ta có thể thấy các cell được đặt khác với trước khi có hard blockage!